

Information, Empathy, and Solidarity in Kindergarten Financing: A Field and Survey Experiment in Duisburg

Achim Goerres Peter John Philipp Chapkovski Jakob Eicheler
Philipp C. Kemper

First draft
May 26, 2026

Abstract

We study whether short empathy and information treatments can strengthen support for solidaristic kindergarten financing. The project combines a field experiment on survey recruitment with a survey experiment on preferences and donation behavior. In Study 1, 13,161 invitation rows were randomized to an efficiency letter or an empathy letter. The empathy letter did not increase take-up: 17.3 percent participated in the empathy condition and 17.6 percent in the efficiency condition, for a difference of -0.24 percentage points, 95 percent CI [-1.06, 1.54], $p = .716$. In Study 2, 2,298 valid survey respondents were assigned to a control group, generic information, an empathy story, or fiscal information about kindergarten costs. The confirmatory treatment contrasts are small and imprecise. No fair-fee or donation contrast survives Holm adjustment. Exploratory analyses point to some differences between the empathy-story and fiscal-information arms in fair-fee preferences and financing satisfaction, and to a relationship between prior belief gaps and donation behavior within the fiscal-information arm. Overall, the paper tells a useful null-result story: in this setting, brief appeals to empathy and fiscal information did not clearly move expressed solidarity. PJ: FEEL IT NEED TO BE A BIT MORE HARD BOILED - INDICATE RESULTS COULD INDICATE DIRECTION FOR FUTURE RESEARCH SAY

Keywords: solidarity; kindergarten fees; childcare; field experiment; survey experiment; empathy; information treatment

1 Introduction

Publicly supported childcare is one of the places where welfare-state solidarity becomes concrete. Kindergarten places are expensive, but parents usually pay only part of the bill; the rest is covered collectively. That makes kindergarten financing a good case for studying how citizens respond to redistribution when the beneficiaries are families and young children. It also connects a very local policy question to broader debates about welfare-state feedback, redistribution, and public opinion (Kumlin and Stadelmann-Steffen, 2014; Kuziemko et al., 2015). Do people become more solidaristic when they learn how a program is financed, or when the beneficiaries are made more vivid through empathy-based communication? NEEDS A BIT OF A LITERATURE AND THEORY HOOK - WE KNOW THIS FROM X, BUT NOT Y THIS DOES Y, MORE OF A NARRATIVE INTRO

This paper reports first results from the READERS WILL NOT KNOW THIS TERM POLIT-SOLID kindergarten experiment in Duisburg. The design has two connected parts. Study 1 asks whether an empathy-oriented invitation letter brings more people into the survey than an efficiency-oriented letter. Study 2 then asks whether respondents change their views or donation behavior after receiving generic information, an empathy story, or fiscal information about kindergarten costs.

The survey treatment arms are as follows: T0 is a control arm with no additional information screen; T1 provides generic information about Duisburg kindergartens and fees; T2 presents an empathy-story vignette about a child benefiting from the solidarity system; and T3 presents fiscal information about taxpayer contributions and the gap between total costs and parental payments. The archived T3 text states that a kindergarten place costs about EUR 993 per month, of which parents pay about EUR 70 and taxpayers cover about EUR 923. This implies a parent share of about 7.05 percent.

The headline result is easy to state TOO INFORMAL: the treatments moved little. The empathy invitation did not increase survey take-up. Within the survey, the treatments produced small and imprecise movements in the fair monthly fee proxy and donation outcome, with no confirmatory contrast surviving Holm correction. Moderator analyses and secondary outcomes do not overturn that conclusion. We therefore frame the paper around an informative null result: brief empathy and information interventions may be insufficient to shift expressed solidarity in this local childcare-finance context. LAST SENTENCE IS GOOD OVERALL THIS NEED TO INTRODUCE THE READER TO THE RESEARCH CHALLENGE WHY IT IS INTERESTING AND HINT WHY A NULL HAS IMPLICATIONS FOR THEORY AND PRACTICE WHILE NOT OVERDOING IT How can we strengthen in-group solidarity in policy communities?

2 Theory and Expectations

The study is built around three broad mechanisms. SO THE STUDY DOES NOT TEST A SPECIFIC MECHANISM -DO OUR FINDINGS SUGGEST N/S THAT EMPATHY IS WHAT MAY BE GOING ON

Information and belief correction. Many citizens may not know how kindergarten places are financed or how large the public subsidy is. Fiscal information could increase support for collective financing by making the solidarity system visible and legitimate. But it could also backfire if the same information makes the taxpayer burden more salient. Existing work on information provision experiments suggests that factual corrections can matter, but that they often depend on prior beliefs, attention, and motivated reasoning (Haaland et al., 2023; Kuziemko et al., 2015).

Empathy and beneficiary visibility. Solidarity may also depend on whether people can picture the beneficiaries. A story about a child benefiting from the kindergarten system could make the policy less abstract, increase perceived deservingness, and strengthen support for solidaristic financing (Batson et al., 1997; van Oorschot, 2000). The field letter tests whether this kind of frame draws people into the study; the survey vignette tests whether it changes what they think or do once they are there.

Heterogeneity. We also expect the treatments to matter more for some respondents than for others. Effects may be stronger among people who are ideologically left-leaning, more educated,

higher-income, or initially more mistaken about the financing split. The analysis uses measures of ideology (v_37), education (v_76), income (v_95), and prior beliefs about the parent share (v_56). ANYTHING HERE POSS TO HAVE TREATMENT EFFECTD AND A NULL AND OUR RESULTS ARE SUGGESTIVE

3 Preregistered Hypotheses

The following hypotheses summarize the preregistered expectations in manuscript language. The final version should still be checked against the exact preregistration wording.

- **Study 1 H1:** The empathy invitation increases survey take-up relative to the efficiency invitation.
- **Study 2 H1a:** The empathy-story and fiscal-information treatments increase the fair monthly fee outcome relative to control.
- **Study 2 H1b:** The empathy-story and fiscal-information treatments increase donation amounts relative to control.
- **Study 2 H2:** Within-survey information effects are stronger after the empathy invitation.
- **Study 2 H3:** Treatment effects are moderated by ideology, with stronger effects among more left-leaning respondents.
- **Study 2 H4–H5:** Treatment effects vary by education and income.
- **Study 2 H6:** The empathy story should move donation behavior more, while fiscal information should move fair-fee preferences more.

4 Data, Treatments, and Outcomes

4.1 Study 1: invitation-letter field experiment

Study 1 contains 13,161 randomized invitation rows. The efficiency-letter condition contains 6,581 rows and the empathy-letter condition contains 6,580 rows. The outcome is survey take-up, measured from the joined invitation randomization files and status-code export.

4.2 Study 2: survey experiment

Study 2 contains 2,298 non-test respondents with valid completion and consent. Treatment assignment is available in c_0001. The four survey arms are:

- **T0:** control, no additional information screen;
- **T1:** generic information about Duisburg kindergartens and fees;

- **T2:** empathy-story vignette about a child benefiting from the solidarity system;
- **T3:** fiscal information about taxpayer contributions and the parent-cost gap.

The fair monthly fee proxy is constructed from `v_65`, `v_67`, and the bracketed current-fee item `v_180`. The donation outcome is recovered from `v_168`; explicit lottery opt-outs in `v_168` or `v_116` are coded as EUR 0 in the main analysis. We also report financing satisfaction, fairness perceptions, confirmed-participant donations, invitation-letter interactions, prior-belief gaps in T3, and moderator analyses.

Table 1: Analysis samples

Sample or observed variable	<i>N</i>
Study 1 randomized invitation rows	13,161
Study 2 valid non-test respondents	2,298
Study 2 respondents with non-missing invitation code	733
Observed fair monthly fee proxy	1,989
Non-missing donation response in <code>v_168</code>	2,198
Explicit lottery opt-out in <code>v_168</code> or <code>v_116</code>	258
Confirmed lottery participation	2,040
Observed prior-belief item <code>v_56</code>	2,003
Observed ideology item	1,852
Observed education item	2,137
Observed income item	2,042

5 Results

5.1 Study 1: empathy invitation and survey take-up

The invitation-letter experiment is very clear TO INFORMAL : the empathy frame did not bring more people into the survey. Take-up was 17.6 percent after the efficiency letter and 17.3 percent after the empathy letter. The estimated empathy-minus-efficiency difference is -0.24 percentage points, 95 percent CI [-1.06, 1.54], $p = .716$. The logistic-regression odds ratio is 0.983.

Table 2: Study 1 invitation-letter take-up

Letter condition	<i>N</i>	Participants	Take-up rate
Efficiency letter	6,581	1,157	0.176
Empathy letter	6,580	1,141	0.173

5.2 Study 2: descriptive arm means

Donation amounts look almost the same across the four survey arms when explicit opt-outs are coded as zero. Fair-fee preferences move around a little more, but not in a clean treatment pattern: the empathy-story arm has the highest mean fair-fee proxy, while the fiscal-information arm has the lowest.

Table 3: Study 2 descriptive means by survey treatment arm

Outcome	Arm	<i>N</i>	Mean	SD
Donation, opt-outs $\rightarrow$ 0	T0	550	27.71	34.98
Donation, opt-outs $\rightarrow$ 0	T1	557	26.82	35.27
Donation, opt-outs $\rightarrow$ 0	T2	591	27.06	34.37
Donation, opt-outs $\rightarrow$ 0	T3	600	27.18	35.20
Fair monthly fee proxy	T0	475	86.50	97.79
Fair monthly fee proxy	T1	482	83.86	102.89
Fair monthly fee proxy	T2	508	90.53	104.67
Fair monthly fee proxy	T3	524	80.24	99.49
Financing satisfaction	T0	497	6.00	3.02
Financing satisfaction	T1	514	6.02	3.04
Financing satisfaction	T2	538	6.29	2.84
Financing satisfaction	T3	531	5.77	2.95

5.3 Confirmatory survey-treatment contrasts

The confirmatory tests compare T1, T2, and T3 to T0 for the fair-fee and donation outcomes, with Holm correction within outcome family. None of the six confirmatory contrasts is statistically decisive after adjustment.

For the fair-fee proxy, T1 is estimated at +EUR 1.65 relative to T0, T2 at +EUR 6.94, and T3 at -EUR 6.37. The confidence intervals all include zero. For donation amounts, the point estimates are -EUR 0.89, -EUR 0.65, and -EUR 0.53 for T1, T2, and T3 respectively. These estimates are small in substantive terms and very imprecise.

Table 4: Confirmatory treatment contrasts

Outcome	Contrast	Estimate	CI low	CI high	Holm <i>p</i>
Fair monthly fee proxy	T1 vs T0	1.65	-9.03	12.33	0.762
Fair monthly fee proxy	T2 vs T0	6.94	-3.61	17.50	0.592
Fair monthly fee proxy	T3 vs T0	-6.37	-16.85	4.11	0.592
Donation, opt-outs $\rightarrow$ 0	T1 vs T0	-0.89	-5.01	3.23	1.000
Donation, opt-outs $\rightarrow$ 0	T2 vs T0	-0.65	-4.71	3.41	1.000
Donation, opt-outs $\rightarrow$ 0	T3 vs T0	-0.53	-4.57	3.52	1.000

5.4 Secondary and exploratory contrasts

The exploratory comparisons produce a few patterns worth keeping in view. The T2–T3 difference in the fair-fee proxy is EUR 13.31, 95 percent CI [3.02, 23.60], $p = .011$. Financing satisfaction is also higher in T2 than T3 by 0.522 points, 95 percent CI [0.166, 0.877], $p = .004$. In contrast, the T2–T3 donation difference is essentially zero (-EUR 0.13, $p = .950$). These results are not the main confirmatory tests, but they raise the possibility that the empathy-story and fiscal-information arms made respondents think about the kindergarten-financing problem in different ways. NEED TO

SAY MORE HERE

Observed donations among confirmed lottery participants also show no meaningful treatment movement: the T1, T2, and T3 contrasts against T0 are -EUR 0.11, +EUR 0.02, and -EUR 0.30 respectively. Financing satisfaction has a suggestive T2 vs T0 contrast (+0.290, $p = .116$) and T3 vs T0 contrast (-0.232, $p = .210$), but these are not decisive.

5.5 Invitation-letter interactions

The treatment-by-letter interaction analysis is only partial because the respondent-level invitation code in `c_0002` is incomplete. The linked sample contains 733 Study 2 respondents. The interaction estimates do not provide clear evidence for the cumulative-empathy hypothesis. The T3-by-empathy-letter interaction is negative for both fair fee (-36.66, $p = .066$) and donation (-12.91, $p = .086$), but this analysis should be read cautiously because the linked sample is small and incomplete.

5.6 Prior-belief gap within T3

The fiscal-information treatment tells respondents that the true parent share is about 7.05 percent. Among T3 respondents with non-missing prior beliefs, the mean estimated parent share is 41.94 percent. The mean absolute belief gap is therefore large, at 35.36 percentage points, with an SD of 23.72.

Within T3, a 10 percentage point larger belief gap is associated with a -EUR 2.57 fair-fee slope ($p = .137$) and a -EUR 1.68 donation slope ($p = .010$). This pattern is interesting, but it should be framed as exploratory and non-causal because the prior-belief gap is not randomized within T3.

5.7 Moderator analyses

The analysis identifies usable moderator items for ideology, education, and income. For ideology, lower values indicate more left-leaning self-placement. Negative treatment-by-ideology coefficients would therefore be directionally consistent with stronger treatment effects among left-leaning respondents. The estimates are directionally suggestive in some cases, especially T3 by ideology for the fair-fee proxy (-10.55, $p = .093$), but they are not robust enough to change the main conclusion. Education and income interactions are also imprecise and do not show a clear pattern. WE MIGHT BEEF THIS UP A BIT

6 Hypothesis-by-Hypothesis Assessment

Table 5: Summary of hypotheses and results

Hypothesis	Verdict	Summary
Study 1 H1: empathy invitation increases take-up	Not supported	Difference = -0.24 pp, 95 percent CI [-1.06, 1.54], $p = .716$; OR = 0.983.
Study 2 H1a: T2/T3 increase fair monthly fee	Not supported	T2 vs T0 = +EUR 6.94, Holm $p = .592$; T3 vs T0 = -EUR 6.37, Holm $p = .592$.
Study 2 H1b: T2/T3 increase donation	Not supported	T2 vs T0 = -EUR 0.65, Holm $p = 1.000$; T3 vs T0 = -EUR 0.53, Holm $p = 1.000$.
Study 2 H2: information effects are stronger after empathy invitation	No clear evidence	Interaction rows are estimable only for the subset with non-missing invitation code.
Study 2 H3: ideology moderates treatment effects	Directional only	Some coefficients are directionally consistent with stronger effects among left-leaning respondents, but evidence is imprecise.
Study 2 H4: education moderates treatment effects	No clear evidence	Education interactions are estimable but imprecise.
Study 2 H5: income moderates treatment effects	No clear evidence	Income interactions do not provide decisive evidence.
Study 2 H6: T2 moves donation more; T3 moves fair fee more	Not supported	Fair-fee T2 vs T3 = EUR 13.31, $p = .011$; donation T2 vs T3 = -EUR 0.13, $p = .950$.

7 Discussion

The evidence points to a null result, but a useful one. Empathy framing did not increase survey take-up in the field experiment, and neither the empathy-story nor fiscal-information survey treatment produced clear confirmatory effects on fair-fee preferences or donation behavior. That matters because the interventions were built around plausible mechanisms from the literature: beneficiary visibility, fiscal-information updating, and solidarity with families using public childcare (Batson et al., 1997; Haaland et al., 2023; van Oorschot, 2000). In this setting, those short messages were not enough.

One possible interpretation is that kindergarten-financing attitudes are relatively stable or only weakly connected to brief survey information. Another is that the treatments activated countervailing considerations. Fiscal information may increase awareness of public support while also making the taxpayer burden salient. Empathy stories may increase concern for beneficiaries among some respondents but produce skepticism or reactance among others. With the current aggregate estimates, we cannot fully separate these possibilities.

The large prior-belief gap in T3 is substantively important. Respondents on average thought parents paid a much larger share of kindergarten costs than the treatment benchmark implied. Yet correcting that context did not produce the expected confirmatory effects. This is useful for theories of information effects because it shows that even large factual gaps do not automatically translate

into attitude or behavior change.

The exploratory T2–T3 differences are best treated as leads for future work rather than as final findings. The empathy story and fiscal-information screen may have moved different considerations, especially around fair-fee judgments and financing satisfaction, but the preregistered contrasts against control do not show reliable treatment effects.

8 Limitations and Next Steps

- Check all hypothesis labels against the preregistration and replace the summary language with exact wording where needed.
- Add more references for childcare politics, local public finance, and German family policy.
- Clarify the construction and interpretation of the fair monthly fee proxy from `v_65`, `v_67`, and `v_180`.
- Report robustness checks for alternative donation codings, including confirmed lottery participants only.
- Add a design figure showing the invitation-letter randomization, survey-treatment randomization, and outcomes.
- Decide how much of the moderator material belongs in the main text versus an appendix.

9 Conclusion

The results provide little evidence that short empathy and information treatments increased solidarity in the Duisburg kindergarten-financing experiment. The empathy invitation did not increase take-up, and the survey treatments did not shift the preregistered fair-fee or donation outcomes after adjustment. The paper can therefore be developed as an informative null result about the limits of light-touch empathy and information interventions in a concrete local welfare-state setting.

References

- Batson, C. D., Polycarpou, M. P., Harmon-Jones, E., Imhoff, H. J., Mitchener, E. C., Bednar, L. L., Klein, T. R., and Highberger, L. (1997). Empathy and attitudes: Can feeling for a member of a stigmatized group improve feelings toward the group? *Journal of Personality and Social Psychology*, 72(1):105–118.
- Haaland, I., Roth, C., and Wohlfart, J. (2023). Designing information provision experiments. *Journal of Economic Literature*, 61(1):3–40.
- Kumlin, S. and Stadelmann-Steffen, I., editors (2014). *How Welfare States Shape the Democratic Public: Policy Feedback, Participation, Voting, and Attitudes*. Edward Elgar Publishing, Cheltenham.

- Kuziemko, I., Norton, M. I., Saez, E., and Stantcheva, S. (2015). How elastic are preferences for redistribution? evidence from randomized survey experiments. *American Economic Review*, 105(4):1478–1508.
- van Oorschot, W. (2000). Who should get what, and why? on deservingness criteria and the conditionality of solidarity among the public. *Policy & Politics*, 28(1):33–48.